

Linear differential equations of second order

A second order diff Eqn is called linear if it can be expressed as

$$\frac{d^2y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = r(x)$$

order = 2
degree = 1

* linear in $\frac{d^2y}{dx^2}$, $\frac{dy}{dx}$ & y

* $p(x)$, $q(x)$, $r(x)$ functions may contain higher powers of x .

* $r(x) = 0$, then it is called homogeneous diff Eqn.

* $r(x) \neq 0$, then it is called Non-homogeneous linear diff Eqn.

Examples & Non-Examples

① $(1-x^2)y'' - 2xy' + 6y = 0$

$$y'' - \frac{2x}{1-x^2}y' + \frac{6}{1-x^2}y = 0$$

Homogeneous
linear diff
Eqn of
second order

② $y'' = \sqrt{y'^2 + 1}$

$$(y'')^2 = y'^2 + 1$$

order = 2
degree = 2

③ $y'' + 4y = e^{-x} \sin x$

Non-homogeneous
linear diff Eqn
of second order.

1. (1) solve $\frac{d^2y}{dx^2} - 4 = 0$

Example ①

solve $\frac{d^2y}{dx^2} - y = 0$

$$\frac{d^2y}{dx^2} = y \quad \text{--- ①}$$

$$y = e^x$$

$$y = e^{-x}$$

$$y' = e^x$$

$$y' = -e^{-x}$$

$$y'' = e^x \rightarrow \text{solution}$$

$$y'' = e^{-x} \rightarrow \text{Another solution}$$

* check that $c_1 e^x + c_2 e^{-x}$ is also a solution of ① for any constant c_1 & c_2 .

$$y = c_1 e^x + c_2 e^{-x}$$

$$y' = c_1 e^x - c_2 e^{-x}$$

$$y'' = c_1 e^x + c_2 e^{-x}$$

Fundamental theorem for homogeneous linear diff equation of second order.

* If y_1 and y_2 are solution of

$$y'' + p(x)y' + q(x)y = 0 \quad \text{--- ①}$$

then any linear combination of y_1 and y_2 are solutions of ①.

Solution:- $\because y_1$ is solution of ①, we have

Solution:- $\because y_1$ is solution of (1)

$$y_1'' + p(x)y_1' + q(x)y_1 = 0$$

$\because y_2$ is solution of (1), we have

$$y_2'' + p(x)y_2' + q(x)y_2 = 0$$

T.P. $Z = C_1 y_1 + C_2 y_2$ is solution of (1)

$$Z' = C_1 y_1' + C_2 y_2'$$

$$Z'' = C_1 y_1'' + C_2 y_2''$$

$$\text{Now } Z'' + p(x)Z' + q(x)Z = C_1 y_1'' + C_2 y_2'' + p(x)(C_1 y_1' + C_2 y_2') + q(x)(C_1 y_1 + C_2 y_2)$$

$$= C_1 (y_1'' + p(x)y_1' + q(x)y_1) + C_2 (y_2'' + p(x)y_2' + q(x)y_2)$$

$$= C_1 \cdot 0 + C_2 \cdot 0 = 0$$

Note:- This theorem is not true for Non-homogeneous linear diff Eqn.

Example:- $\frac{d^2 y}{dx^2} + y = 1$ ——— (★)

$$y_1 = \sin x + 1$$

$$y_1' = \cos x$$

$$y_1'' = -\sin x$$

$$y_2 = \cos x + 1$$

$$y_2' = -\sin x$$

$$y_2'' = -\cos x$$

Is $2 + \sin x + \cos x$ solution of (★)

$$z = 2 + \sin x + \cos x$$

$$z' = \cos x - \sin x$$

$$z'' = -\sin x - \cos x$$

$$z'' + z = 2$$

It is not solution of $\odot$

Note:- The theorem is not applicable to Non-linear diff equations.

Example:- $y \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0 \quad \text{--- (2)}$

- $y=1$ is solution of (2)
- $y=x^2$ is solution of (2)

Check $z = x^2 + 1$ is not solution of (2)

$$z' = 2x \quad z'' = 2$$

$$(x^2 + 1)2 - x(2x) \neq 0$$

General solution:- No. of constants = order of diff Eqn.

- If diff Eqn is of second order, then its complete/general solution of type

$$y = C_1 y_1 + C_2 y_2, \quad C_1, C_2 \text{ are constant}$$

- y_1 and y_2 must be L.I. function, they are called Basis of solution set of second order diff Eqn.

they are called second order diff Eqn.

Example:- Initial value problems in second order

$$y'' + p(x)y' + q(x)y = 0$$

$$y(x_0) = y_0, \quad y'(x_0) = y'_0$$

↓
Boundary condition

↪ Boundary value problem.

Example:- $y'' - y = 0$, $y(0) = 4$, $y'(0) = 1$

Complete solution:- $y = c_1 e^x + c_2 e^{-x}$

$$y(0) = 4 \Rightarrow 4 = c_1 + c_2$$

$$y'(0) = 1 \Rightarrow y' = c_1 e^x - c_2 e^{-x}$$

$$1 = c_1 - c_2$$

$$c_1 = \frac{5}{2}$$

$$c_2 = \frac{3}{2}$$

Particular solution:- $y = \frac{5}{2} e^x + \frac{3}{2} e^{-x}$

* If we take $y_2 = 2e^x$

$$y = c_1 e^x + 2c_2 e^x$$

$$y(0) = 4 \Rightarrow$$

$$4 = c_1 + 2c_2$$

$$y'(0) = 1 \Rightarrow$$

$$1 = c_1 + 2c_2$$

} No solution

1 1 solution of homogeneous linear

* To find solution of homogeneous linear differential Eqn of second order, we need to find two L.I. solutions of diff Eqn.

Reduction of order method

• Homogeneous linear diff Eqn of second order

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0 \quad \text{--- (1)}$$

we know that y_1 is solution eqn (1),

$$y_1'' + p(x)y_1' + q(x)y_1 = 0 \quad \text{--- (2)}$$

we assume $y_2 = u(x)y_1$ is also a solution of (1)

$$y_2' = u(x)y_1' + u'(x)y_1$$

$$y_2'' = u(x)y_1'' + 2u'(x)y_1' + u''(x)y_1$$

$\therefore y_2$ is solution of (1), we have

$$y_2'' + p(x)y_2' + q(x)y_2 = 0$$

$$u(x)y_1'' + 2u'(x)y_1' + u''(x)y_1 + p(x)(u(x)y_1' + u'(x)y_1) + q(x)(u(x)y_1) = 0$$

$$u(x)(y_1'' + p(x)y_1' + q(x)y_1) + u'(x)(2y_1' + p(x)y_1) + u''(x)y_1 = 0$$

$$\Rightarrow u'(x)(2y_1' + p(x)y_1) + u''(x)y_1 = 0$$

$$\Rightarrow u'(x) (2y_1' + p(x)y_1) + u''(x)y_1 = 0$$

Substitute $u'(x) = \varphi(x)$

we get $\varphi(x) (2y_1' + p(x)y_1) + \varphi'(x)y_1 = 0$

$$\varphi (2y_1' + p(x)y_1) + \frac{d\varphi}{dx} y_1 = 0$$

$$\varphi (2y_1' + p(x)y_1) = -\frac{d\varphi}{dx} y_1$$

$$\Rightarrow \frac{(-2y_1' - p(x)y_1)}{y_1} dx = \frac{d\varphi}{\varphi}$$

$$\Rightarrow \int \frac{d\varphi}{\varphi} = \int -2 \frac{y_1'}{y_1} dx - \int p(x) dx$$

$$\ln|\varphi| = -2 \ln|y_1| - \int p(x) dx$$

$$u'(x) = \varphi = y_1^{-2} e^{-\int p(x) dx}$$

$$u(x) = \int \varphi dx = \int y_1^{-2} e^{-\int p(x) dx} dx$$

function of x

$$y_2 = u(x) y_1$$

y_2 is L.S. of y_1 .

Question:-

$$y'' = y'$$

$y_1 = 1$ is solution.

$$y_2 = u(x) \cdot 1 = u(x)$$

we get

$$u''(x) = u'(x)$$

put $\varphi(x) = u'(x)$, we get $\varphi'(x) = \varphi(x)$

$\varphi(x)$ is solution of diff Eqn $z' = z$

$$\frac{dz}{dx} = z \Rightarrow \frac{dz}{z} = dx$$

$$\Rightarrow \ln|z| = x \Rightarrow z = e^x$$

$$u'(x) = \varphi(x) = e^x \Rightarrow u(x) = \int \varphi(x) dx = \int e^x dx = e^x$$

$$y_2 = u(x)y_1 = e^x \cdot 1 = e^x$$

Complete solution :- $y = C_1 \cdot 1 + C_2 e^x$

* $y'' = y'$, Let $y_1 = e^x$

$$y_2 = u(x)e^x$$

$$y_2' = u(x)e^x + u'(x)e^x$$

$$y_2'' = u(x)e^x + 2u'(x)e^x + u''(x)e^x$$

$$\cancel{u(x)e^x} + 2u'(x)e^x + u''(x)e^x = \cancel{u(x)e^x} + u'(x)e^x$$

$$(u'(x) + u''(x))(e^x) = 0$$

$$\Rightarrow u'(x) + u''(x) = 0$$

substitute

$$\varphi(x) = u'(x) \Rightarrow \varphi'(x) + \varphi(x) = 0$$

$\Rightarrow \varphi(x)$ is solution of diff Eqn $z' + z = 0$

$\Rightarrow \varphi(x)$ is solution of diff eqn $z' + z = 0$

$$\frac{dz}{dx} = -z \Rightarrow \frac{dz}{z} = -dx$$

$$\Rightarrow \ln|z| = -x$$

$$\Rightarrow z = e^{-x}$$

$$\Rightarrow u'(x) = \varphi(x) = e^{-x} \Rightarrow u(x) = \int \varphi(x) dx = \int e^{-x} dx$$

$$\Rightarrow u(x) = -e^{-x}$$

$$y_2 = u(x)y_1 = -e^{-x} \cdot e^x = -1$$

Complete solution $y = C_1 e^x + C_2 (-1)$.

* Linearly dependent functions :- Two functions $\varphi_1(x)$ and $\varphi_2(x)$ defined on some domain D are said to be linearly dependent if $\exists$ non-zero constants C_1 and C_2 s.t.

$$C_1 \varphi_1(x) + C_2 \varphi_2(x) = 0 \quad \forall x \in D,$$

• Linearly independent functions :- If

$$\forall x \in D, \quad C_1 \varphi_1(x) + C_2 \varphi_2(x) = 0 \Rightarrow C_1 = 0 \text{ \& } C_2 = 0$$

then we say that φ_1 and φ_2 are linearly independent functions.

Wronskian :- Given any two functions φ_1 and φ_2 , Wronskian of φ_1 and φ_2 is defined as

$$W(\varphi_1, \varphi_2) = \begin{vmatrix} \varphi_1 & \varphi_2 \\ \varphi_1' & \varphi_2' \end{vmatrix}$$

- Two functions φ_1 and φ_2 are linearly independent on Domain D iff $W(\varphi_1, \varphi_2) \neq 0$ on Domain D .

$$* \quad W(\varphi_1, \varphi_2, \varphi_3) = \begin{vmatrix} \varphi_1 & \varphi_2 & \varphi_3 \\ \varphi_1' & \varphi_2' & \varphi_3' \\ \varphi_1'' & \varphi_2'' & \varphi_3'' \end{vmatrix}$$

- * Example:- show that e^x and e^{2x} are L.I. $\forall x \in [3, 4]$

$$W(e^x, e^{2x}) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{2x}e^x - e^{2x}e^x = e^{3x} \neq 0 \quad \forall x \in [3, 4]$$

$\Rightarrow e^x$ and e^{2x} are L.I. on $[3, 4]$.

- * Diff Eqn of first order and Higher degree

General form

$$f(x, y, p) = 0,$$

$$p = \frac{dy}{dx}$$

Equations solvable for p

$\Downarrow$
polynomial eqns in p

Equns solvable for y

$$\Downarrow$$

 $y = f(x, p)$

$$dy = \left[\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial p} dp \right]$$

Eqns solvable for x

$$\Downarrow$$

 $x = f(y, p)$
 $\Downarrow$

$$\frac{dx}{\dots} = \left[\frac{1}{\dots} = \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial p} dp \right]$$

1 0 in p

$$\frac{dy}{dx} = p = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial p} dp$$

$$\frac{dx}{dy} = \frac{1}{p} = \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial p} dp$$

Example :-

$$xp^2 + x = 2yp$$

$$2y = \frac{xp^2 + x}{p}$$

$$2y = xp + \frac{x}{p}$$

diff w.r.t x, we get $2 \frac{dy}{dx} = p + x \frac{dp}{dx} + \frac{1}{p} - \frac{x}{p^2} \frac{dp}{dx}$

$$\Rightarrow 2p = p + x \frac{dp}{dx} + \frac{1}{p} - \frac{x}{p^2} \frac{dp}{dx}$$

$$\Rightarrow p - x \frac{dp}{dx} - \frac{1}{p} + \frac{x}{p^2} \frac{dp}{dx} = 0$$

$$\Rightarrow p^3 - p^2 x \frac{dp}{dx} - p + x \frac{dp}{dx} = 0$$

$$\Rightarrow p(p^2 - 1) - x \frac{dp}{dx} (p^2 - 1) = 0$$

$$\Rightarrow \left(p - x \frac{dp}{dx} \right) (p^2 - 1) = 0$$

$$p^2 - 1 = 0$$

$$p - x \frac{dp}{dx} = 0$$

$$p - x \frac{dp}{dx} = 0$$

$$p = x \frac{dp}{dx}$$

$$\frac{dx}{x} = \frac{dp}{p} \Rightarrow p = xc$$

$$p^2 - 1 = 0$$

$$p^2 = 1$$

$$p = 1$$

$$p = -1$$

Solution is

$$x(xc)^2 + x = 2y(xc)$$

$$x^2 c^2 + 1 = 2yc$$

General solution

$$2x = 2y$$

$$x = y$$

singular

$$2x = -2y$$

$$x = -y$$

solution

Example :-

$$x^2 p^2 + xyp - 6y^2 = 0$$

$$x^2 p^2 + 3xyp - 2xyp - 6y^2 = 0$$

$$xp(xp + 3y) - 2y(xp + 3y) = 0$$

$$(xp - 2y)(xp + 3y) = 0$$

$$xp - 2y = 0$$

$$x \frac{dy}{dx} = 2y$$

$$\frac{dy}{y} = 2 \frac{dx}{x}$$

$$\Rightarrow y = x^2 c$$

$$xp + 3y = 0$$

$$x \frac{dy}{dx} = -3y$$

$$\frac{dy}{y} = -3 \frac{dx}{x}$$

$$\Rightarrow y = x^{-3} c'$$

Example :-

$$y = 2px + y^2 p^3$$

$$y - y^2 p^3 = 2px$$

$$\Rightarrow \frac{y}{p} - y^2 p^2 = 2x$$

Diff w.r.t y

$$\Rightarrow \frac{1}{p} - \frac{y}{p^2} \cdot \frac{dp}{dy} - 2yp^2 - 2y^2 p \frac{dp}{dy} = 2 \frac{dx}{dy} = \frac{2}{p}$$

$$\Rightarrow p - y \frac{dp}{dy} - 2yp^4 - 2y^2p^3 \frac{dp}{dy} - 2p = 0$$

$$\Rightarrow p + y \frac{dp}{dy} + 2yp^4 + 2y^2p^3 \frac{dp}{dy} = 0$$

$$\Rightarrow p(1 + 2yp^3) + y \frac{dp}{dy} (1 + 2yp^3) = 0$$

$$\Rightarrow \left(p + y \frac{dp}{dy} \right) (1 + 2yp^3) = 0$$

$$p + y \frac{dp}{dy} = 0$$

$$\frac{dp}{p} = -\frac{dy}{y}$$

$$\boxed{p = C/y}$$

$$y = \frac{2xc}{y} + y^2 \frac{C^3}{y^3}$$

$$y^2 = 2xc + C^3$$

Complete solution

$$1 + 2yp^3 = 0$$

$$2yp^3 = -1$$

$$p^3 = -\frac{1}{2y} \Rightarrow p = \left(-\frac{1}{2y} \right)^{1/3}$$

$$y = 2x \left(-\frac{1}{2y} \right)^{1/3} + y^2 \cdot \frac{-1}{2y}$$

singular solution